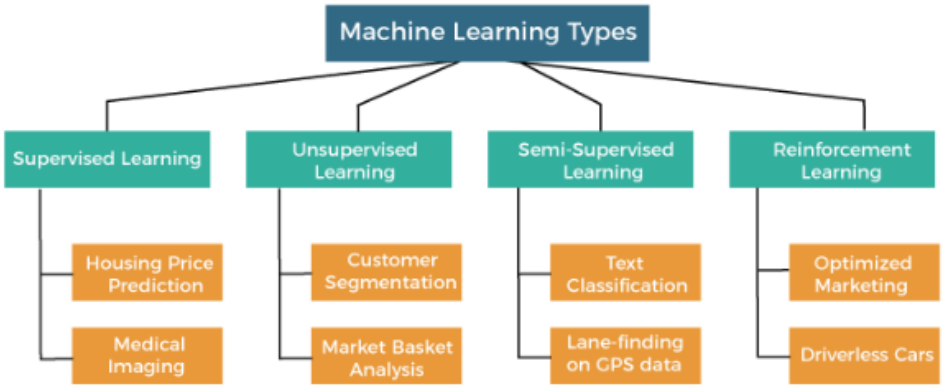
1. Types of Machine Learning

Machine Learning teaches systems to learn patterns and make decisions like humans by analyzing and learning from data.

There are several types of machine learning, each with special characteristics and applications. Some of the main types of machine learning algorithms are as follows:

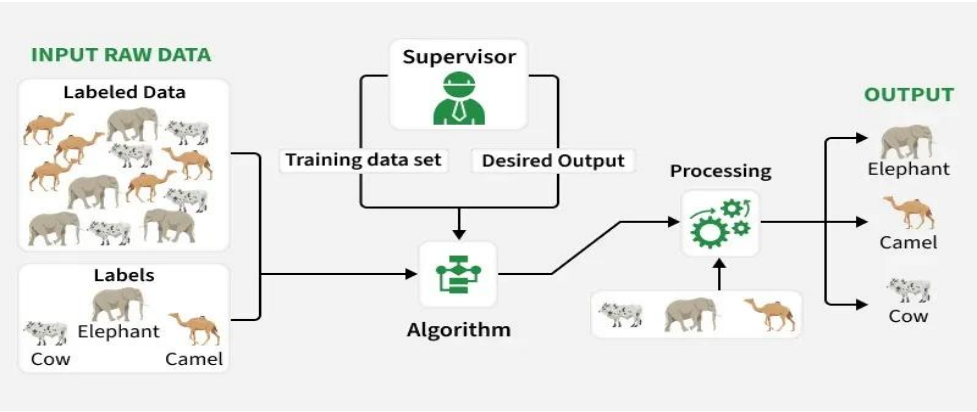
1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Reinforcement Learning
4. Semi-Supervised Learning

Additionally, there is a more specific category called Semi-Supervised Learning and Self-Supervised Learning, which combines elements of both supervised and unsupervised learning



1. Supervised Machine Learning

Supervised learning is defined as when a model gets trained on a "Labeled Dataset". Labelled datasets have both input and output parameters. In Supervised Learning algorithms learn to map points between inputs and correct outputs. It has both training and validation datasets labelled.



Example: If you train a model using labeled images of cats and dogs, it learns the features of each. When shown a new image, it predicts whether it's a cat or a dog.
There are two main categories of supervised learning that are mentioned below:

1. Classification

Classification predicts categorical outputs, meaning it assigns data into predefined classes like spam/non-spam emails or disease risk categories. These algorithms learn to map input features to discrete labels. Here are some classification algorithms:

Common Classification Algorithms

1. Logistic Regression

A statistical algorithm used for binary classification problems. It estimates the probability that an input belongs to a particular class.

Example: Predicting whether a customer will churn (Yes/No).

Advantages:

Simple and fast

Easy to interpret

Works well for linearly separable data

2. Decision Tree

A tree-like model that makes decisions by splitting data based on feature values.

Example: Loan approval based on income, age, and credit score.

Advantages:

Easy to visualize and understand

Handles both numerical and categorical data

Requires little data preprocessing

3. Random Forest

An ensemble method that combines multiple decision trees and uses majority voting for classification.

Example: Fraud detection in banking transactions.

Advantages:

Higher accuracy than a single decision tree

Reduces overfitting

Handles large datasets well

4. K-Nearest Neighbors (KNN)

Classifies a data point based on the classes of its nearest neighbors.

Example: Classifying a handwritten digit based on similar examples.

Advantages:

Simple to implement

No training phase required

Effective for small datasets

5. Naive Bayes

A probabilistic classifier based on Bayes' Theorem that assumes features are independent.

Example: Email spam filtering.

Advantages:

Fast and efficient

Works well with text data

Performs well even with limited training data

6. Support Vector Machine (SVM)

Finds the optimal boundary (hyperplane) that best separates classes.

Example: Image classification (cat vs. dog).

Advantages:

Effective in high-dimensional datasets

Works well with complex decision boundaries

Good performance on smaller datasets

Summary Table

Algorithm	Best For	Key Advantage
Logistic Regression	Binary classification	Simple and interpretable
Decision Tree	Rule-based decisions	Easy to understand
Random Forest	Complex datasets	High accuracy, less overfitting
KNN	Similarity-based classification	Simple and intuitive
Naive Bayes	Text classification	Fast and efficient
SVM	High-dimensional data	Strong classification performance

2. Regression

Regression, predicts continuous values, such as house prices or product sales. It learns the relationship between input features and a numerical target variable. Here are some regression algorithms:

Common Regression Algorithms

1. Linear Regression

The simplest regression algorithm that models a linear relationship between input features and the target variable.

Example: Predicting house prices based on square footage.

Advantages:

- Easy to understand and implement
- Fast training and prediction
- Highly interpretable

2. Polynomial Regression

An extension of Linear Regression that can model curved (non-linear) relationships by adding polynomial terms.

Example: Predicting crop yield based on rainfall, where the relationship is not linear.

Advantages:

- Captures non-linear patterns
- More flexible than Linear Regression

3. Ridge Regression

A regularized version of Linear Regression that adds a penalty term to reduce overfitting.

Example: Predicting sales when many features are highly correlated.

Advantages:

- Reduces overfitting
- Handles multicollinearity well
- Improves model stability

4. Lasso Regression

Similar to Ridge Regression but can reduce some feature coefficients to zero, effectively performing feature selection.

Example: Identifying the most important factors affecting house prices.

Advantages:

- Reduces overfitting
- Automatically selects important features
- Produces simpler models

5. Decision Tree Regression

Uses a tree structure to split data into regions and predict values based on those splits.

Example: Predicting car prices using factors such as age, mileage, and brand.

Advantages:

- Handles non-linear relationships
- Easy to visualize
- Requires little data preprocessing

6. Random Forest Regression

An ensemble of multiple decision trees whose predictions are averaged to produce a final result.

Example: Predicting future sales from historical business data.

Advantages:

- High accuracy
- Handles complex relationships
- Less prone to overfitting than a single decision tree

Summary Table

Algorithm	**Best For**	**Key Advantage**
Linear Regression	Linear relationships	Simple and interpretable
Polynomial Regressi	Curved relationships	Captures non-linearity
Ridge Regression	Many correlated features	Reduces overfitting
Lasso Regression	Feature selection	Removes unimportant fe
Decision Tree Regres	Non-linear data	Easy to understand
Random Forest Regr	Complex datasets	High accuracy and robu

Key Difference: Classification vs Regression

Classification	**Regression**
Predicts categories	Predicts numerical values
Output: Spam/Not Spam	Output: House Price = $250,000
Uses Accuracy, Precision, Recall, I	Uses MAE, MSE, RMSE, R^2

Where to Use Supervised Learning

- When you have labeled data and want to predict outcomes.

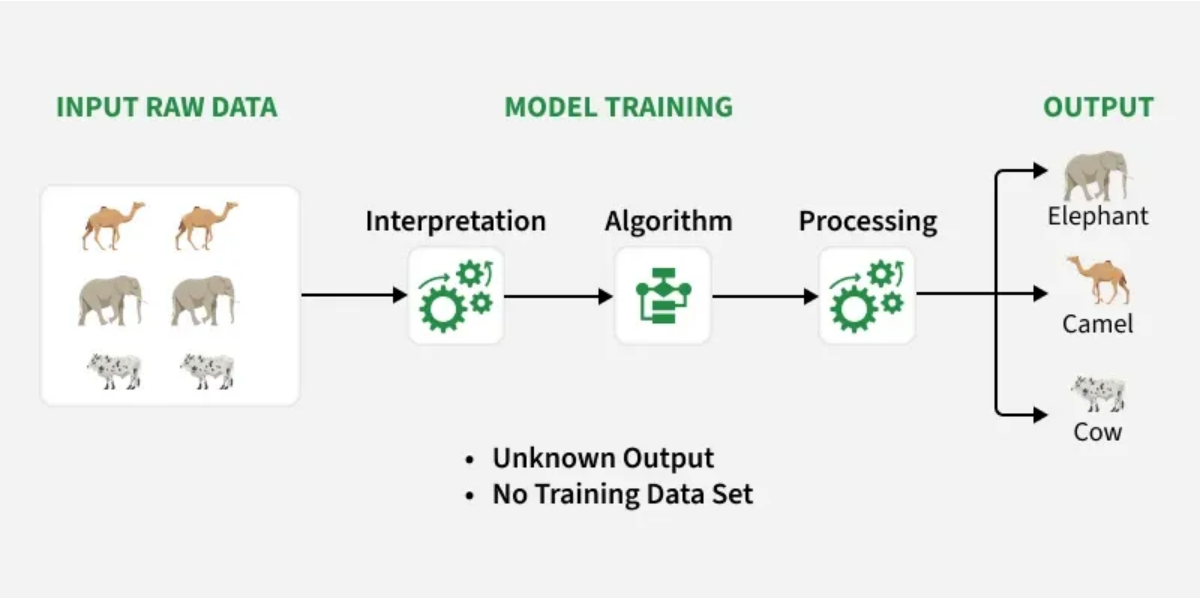
- Ideal for classification (like spam detection) or regression tasks (like price forecasting).
- Best used in domains where historical data with outcomes is already available.

Applications: Supervised learning is used in a wide variety of applications, including:

- **Image, speech and text processing:** For tasks like image classification, speech recognition and sentiment analysis.
- **Predictive analytics:** To forecast sales, customer churn, stock prices and weather conditions.
- **Recommendation and personalization:** Powering systems that suggest products, movies or content.
- **Healthcare and finance:** Used for medical diagnosis, fraud detection and credit scoring.
- **Automation and control:** In autonomous vehicles, manufacturing quality checks and gaming AI.

2. Unsupervised Machine Learning

Unsupervised Learning works with unlabeled data, meaning there are no predefined outputs. The algorithm finds hidden patterns, groups or relationships within the data on its own. It's mainly used for clustering, dimensionality reduction and data visualization.



Example: If you have customer data without labels, the algorithm can group similar customers based on purchase behavior useful for segmentation and marketing.

There are two main categories of unsupervised learning that are mentioned below:

1. Clustering

Clustering is the process of grouping data points into clusters based on their similarity. This technique is useful for identifying patterns and relationships in data without the need for labeled examples. Common techniques include:

1. K-Means

K-Means divides data into **K predefined clusters**. Each data point is assigned to the cluster with the nearest centroid (center point).

Algorithm:

1. Choose the number of clusters (K).
2. Randomly initialize centroids.
3. Assign points to the nearest centroid.
4. Recalculate centroids.
5. Repeat until clusters stabilize.

Example:

Grouping customers into:

- Budget buyers
- Regular buyers
- Premium buyers

Advantages:

- Simple and fast
- Works well on large datasets
- Easy to implement

Limitation:

- Must specify the number of clusters (K) beforehand.
- Sensitive to outliers.

2. DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

DBSCAN groups points based on **density**. Areas with many nearby points form clusters, while isolated points are treated as noise (outliers).

Algorithm:

- Finds dense regions of data.
- Expands clusters from dense points.
- Identifies sparse points as outliers.

Example:

Detecting fraudulent transactions or unusual customer behavior.

Advantages:

- Does not require specifying the number of clusters.
- Detects outliers automatically.
- Handles irregularly shaped clusters.

Limitation:

- Performance depends on choosing appropriate density parameters.

3. Mean Shift

Mean Shift identifies clusters by moving data points toward regions with the highest data density.

Algorithm:

1. Place a window around data points.
2. Compute the mean of points inside the window.
3. Shift the window toward the mean.
4. Repeat until convergence.
5. Points converging to the same location form a cluster.

Example:

Image segmentation and object tracking in computer vision.

Advantages:

- No need to specify the number of clusters.
- Can discover clusters of arbitrary shapes.
- Finds cluster centers automatically.

Limitation:

- Computationally expensive for large datasets.

Summary Table

Algorithm	Requires Numb Clusters?	Handles Outli	Best For
K-Means	Yes (K)	No	Large datasets with well-se clusters
DBSCAN	No	Yes	Finding arbitrary-shaped cluste anomalies
Mean Shift	No	Partially	Discovering unknown cluster struc

2. Dimensionality Reduction Techniques

Dimensionality reduction helps reduce the number of features while preserving important information. Common techniques include:

- Principal Component Analysis
- Independent Component Analysis

3. Association Rule Learning

Association rule learning is a technique for discovering relationships between items in a dataset. It identifies rules that indicate the presence of one item implies the presence of another item with a specific probability. Common techniques include:

- Apriori
- FP-growth
- Eclat

Where to Use Unsupervised Learning

- When data is unlabeled or unstructured.
- Useful for exploratory analysis, clustering or feature extraction.
- Common in marketing, recommendation systems and fraud detection where patterns matter more than labels.

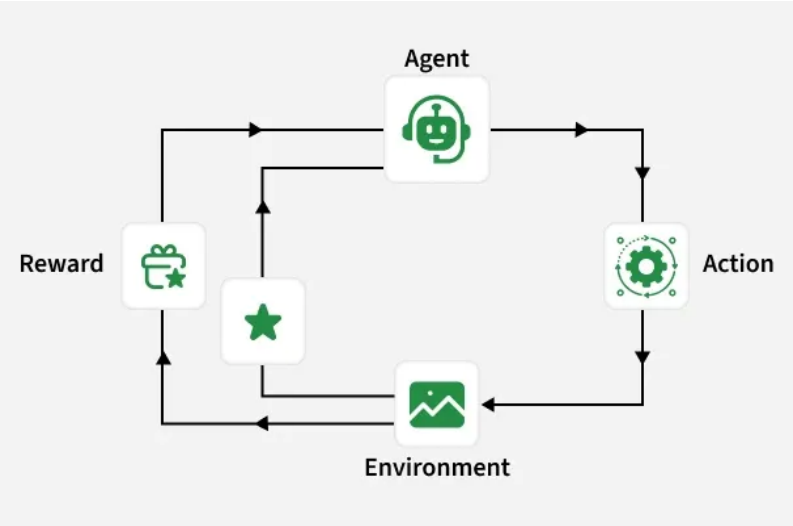
Applications of Unsupervised Learning

Here are some common applications of unsupervised learning:

- **Clustering and segmentation:** Group similar data points, customers or images.
- **Anomaly detection:** Spot unusual patterns or outliers in data.
- **Dimensionality reduction:** Simplify large datasets while retaining key information.
- **Recommendation and marketing:** Identify user preferences and improve product suggestions.
- **Data preprocessing and analysis:** Clean data, detect patterns and support exploratory data analysis (EDA).

3. Reinforcement Learning

Reinforcement learning trains an agent to make a sequence of decisions through trial and error. The agent interacts with the environment, receives feedback in the form of rewards or penalties and learns optimal actions over time.



Example: An AI agent learning to play chess gets positive feedback for good moves and negative for poor ones. Over time, it learns strategies to win more often.

Here are some of most common reinforcement learning algorithms:

- **Q-learning:** Learns the best action for each state based on expected rewards.
- **SARSA (State-Action-Reward-State-Action):** Similar to Q-learning but updates values for the action actually taken.
- **Deep Q-learning:** Uses neural networks to handle complex state-action relationships

Types of Reinforcement Learning

- **Positive Reinforcement:** Rewards desired behavior (e.g., giving points for correct answers).
- **Negative Reinforcement:** Removes negative outcomes to encourage good actions (e.g., turning off a buzzer after the right move).

Where to Use Reinforcement Learning

- When you need an agent to learn by interacting with an environment.
- Best for decision-making or optimization tasks involving trial and feedback loops.
- Used when long-term performance or adaptive behavior is more important than immediate accuracy.

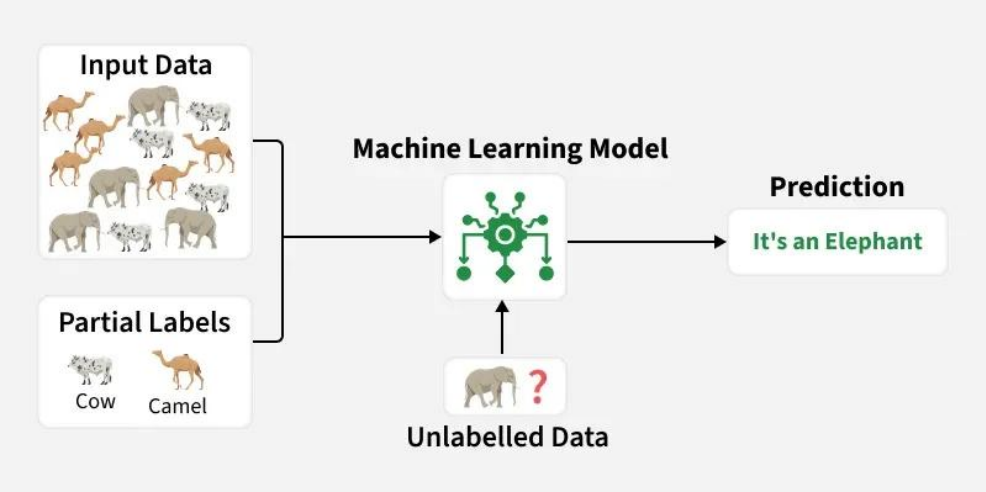
Applications of Reinforcement Learning

Here are some applications of reinforcement learning:

- **Gaming and simulation:** Teaching agents or NPCs to play and adapt intelligently.
- **Robotics and automation:** Enabling robots to perform tasks autonomously.
- **Autonomous vehicles:** Helping self-driving cars make real-time decisions.
- **Healthcare and finance:** Optimizing treatment plans, trading and resource allocation.
- **Recommendation and personalization:** Improving user experience through adaptive suggestions.
- **Industrial and energy management:** Optimizing control systems and energy use.

Semi-Supervised Learning: Supervised + Unsupervised Learning

Semi-Supervised learning Semi-Supervised Learning combines both Supervised and Unsupervised approaches. It uses a small set of labeled data and a large set of unlabeled data for training useful when labeling is costly or time-consuming.



Example: Consider that we are building a language translation model, having labeled translations for every sentence pair can be resource intensive. It allows the models to learn from labeled and unlabeled sentence pairs, making them more accurate. This technique has led to significant improvements in the quality of machine translation services.

Popular Techniques

- **Graph-based Learning:** Spreads label information through data relationships.
- **Label Propagation:** Iteratively assigns labels to unlabeled data.
- **Co-training:** Uses two models to train and label each other's data.
- **Self-training:** Uses model predictions as pseudo-labels.
- **Generative Adversarial Networks (GANs):** Generates synthetic data to improve learning.

Where to Use Semi-Supervised Learning

- When you have limited labeled data but plenty of unlabeled data.
- Useful for domains with high labeling costs, such as medical, NLP or image datasets.
- Ideal when unlabeled data still holds valuable information that can improve learning performance.

Applications

- **Image Classification:** Combine small labeled and large unlabeled image datasets to improve accuracy.
- **Natural Language Processing (NLP):** Enhance language models by using a mix of labeled and vast unlabeled text data.

- **Speech Recognition:** Boost accuracy by leveraging limited transcribed audio and more unlabeled speech data.
- **Recommendation Systems:** Improve recommendations using sparse labeled data and abundant unlabeled user behavior.
- **Healthcare & Medical Imaging:** Improve medical image analysis with a mix of labeled and unlabeled images.

Self-Supervised Learning

Self-Supervised Learning (SSL) is a modern approach where models generate their own labels from raw data. It doesn't rely on manual annotation; instead, the model learns by predicting parts of data from other parts.

Example: In NLP, models like BERT or GPT learn by predicting masked words in sentences, using surrounding context as supervision. This helps them learn language understanding without human labeling.

Popular Techniques

- Masked Modeling (BERT)
- Contrastive Learning (SimCLR, MoCo)
- Autoencoders
- Predictive Coding

Applications

- Natural Language Processing
- Computer Vision and Speech Recognition
- Video understanding
- Pre-training for large AI models

Where to Use Self-Supervised Learning

- When manual labeling is impossible or expensive.
- Suitable for large-scale datasets like text, audio and images.
- Best for pre-training models that can later be fine-tuned for specific supervised tasks.

Type	**Data Require**	Label Availabili	Learning Goal	**Common Use Case**
Supervised	High	Labeled	Predict outputs	Spam detection, prediction
Unsupervised	Medium	Unlabeled	Find hidden pat	Customer segmentat
Reinforcement	High	Reward feedbac	Learn best actio	Robotics, Games
Semi-Supervised	Medium	Partial labels	Combine both le types	NLP, Image recognit
Self-Supervised	High	Self-generated la	Learn representations	BERT, GPT, CLIP